

TC39x BD-Step

Customer documentation update information for TC39x BD-Step



ISO 26262
compliant

About this document

Scope and purpose

Infineon provided to customer a Safety Case Report for the TC39x BD-Step Microcontroller which has been amended by Infineon by respective addenda provided by Infineon to customer (collectively “Safety Case Report”). Please note that, irrespective of any information to the contrary set forth in the Safety Case Report, the most recent version of any Customer Documentation as listed in Table 9 “Reference List of Customer Documentation and Internal Documents” of the Safety Case Report, which is applicable to the TC39x BD-Step Microcontroller is as stated in the latest Safety Package Release Note provided by Infineon to customer.

All relevant changes on the updated documentation have been assessed and necessary modification have been carried out as per the standard change and configuration management practice. Application of standard change and configuration management procedures ensure, that the functional safety of the developed product is not affected by the applied changes to the listed documents.

The initial claim for a compliant development according to ISO 26262:2011 is not affected by any of the applied document changes.

Customers shall use documents as listed in the corresponding Safety Package Release Note marked as Safety Related Update “Y” for their respective applications.

Customers are recommended to use the documents as listed in the corresponding Safety Package Release Note marked as Safety Related Update “N” for their respective applications.

If the product is intended to be used in a functional safety application, you are obliged to ensure conformity with the relevant requirements of ISO 26262 [1].

In particular you are obliged to obtain the available product safety documents and to respect the information and information updates provided by Infineon Technologies AG.

In the event that you merely resell the product as a distributor, you are obliged to ensure that your customers obtain the available product safety documents and that your customers respect the information and information updates provided by Infineon.

Information updates are provided via Product Change Notification (PCN), Info Note or Early Problem Notification (EPN) or any other written communication to customers or distributors that have ordered the product within the last 5 years. The complete updated safety document is available via the Infineon sales office.

Intended audience

This document is written for experienced project managers, system integrators and functional safety managers involved in the development of a safety-related system who consider integrating the TC39x BD-Step into their application.

About this document**Scope**

For the list of applicable variants and derivatives see the respective table [Table 2].



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1 Functional Safety product documentation

The customer Safety Case documents which are applicable for this document are listed in the table below [Table 1].

Table 1

Document Title	Revision Number	Release Date
AURIX™ TC39x BC-Step Microcontroller Safety Case Report [R1.0]	V1.00	2019-01
AURIX™ TC39x BD-Step Microcontroller Safety Case Addendum; [A1.0]	V1.0	2020-03

This document applies to the TC39x BD-Step Microcontroller product variants given in the table below [Table 2].

Table 2 TC39x BD-Step Microcontroller product variants and derivatives

Name	Description and features	Package	Marking
TC39x BD	-40 °C to 150 °C; Hexa Core Emulation Device; 16 MB Flash; 300 MHz	PG-LFBGA-516	SAL-TC399XX-256F300S BD
	-40 °C to 150 °C; Hexa Core Emulation Device; 16 MB Flash; 300 MHz		SAL-TC399XP-256F300S BD
	-40 °C to 125 °C; Hexa Core Emulation Device; 16 MB Flash; 300 MHz		SAK-TC399XP-256F300S BD
	-40 °C to 125 °C; Hexa Core Emulation Device; 16 MB Flash; 300 MHz		SAK-TC399XX-256F300S BD
	-40 °C to 150 °C; Hexa Core Emulation Device; 16 MB Flash; 300 MHz	PG-LFBGA-292	SAL-TC397XP-256F300S BD
	-40 °C to 125 °C; Hexa Core Emulation Device; 16 MB Flash; 300 MHz		SAK-TC397XP-256F300S BD
	-40 °C to 125 °C; Hexa Core Emulation Device; 16 MB Flash; 300 MHz		SAK-TC397XA-256F300S BD
	-40 °C to 125 °C; Quad Core Emulation Device; 10 MB Flash; 300 MHz		SAK-TC397QA-160F300S BD
	-40 °C to 125 °C; Hexa Core Emulation Device; 16 MB Flash; 300 MHz		SAK-TC397XX-256F300S BD
	-40 °C to 125 °C; Quad Core Emulation Device; 12 MB Flash; 300 MHz		SAK-TC397QP-192F300S BD

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	-40 °C to 125 °C; Quad Core Emulation Device; 16 MB Flash; 300 MHz		SAK-TC397QP-256F300S BD
	-40 °C to 125 °C; Hexa Core Emulation Device; 16 MB Flash; 300 MHz		SAK-TC397XZ-256F300S BD
	-40 °C to 125 °C; Hexa Core Emulation Device; 16 MB Flash; 300 MHz		SAK-TC397XM-256F300S BD
	-40 °C to 150 °C; Quad Core Emulation Device; 12 MB Flash; 300 MHz		SAL-TC397QP-192F300S BD
	-40 °C to 150 °C; Quad Core Emulation Device; 16 MB Flash; 300 MHz		SAL-TC397QP-256F300S BD
	-40 °C to 150 °C; Hexa Core Emulation Device; 16 MB Flash; 300 MHz		SAL-TC397XZ-256F300S BD
	-40 °C to 150 °C; Hexa Core Emulation Device; 16 MB Flash; 300 MHz		SAL-TC397XX-256F300S BD

This document is applicable for customer relevant documentation as listed in [Table 3]. The table lists all relevant documents with their initial information regarding revision number(s) and corresponding release date(s).

The last valid revision number and corresponding release date is always listed within the Safety Package Release Note document which is contained within a released Safety Package for a particular set of product variants.

If different revisions are listed for the very same document in the table below, a reference is given for each different version pointing to the corresponding Safety Case document, which is either the initial Safety Case Report or a following Safety Case Addendum.

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Table 3 List of Customer Documentation

Document	Initial Revision Number & Release Date
AURIX™ TC3xx IFX User Manual	V1.0.0; 2018-08; [R1.0] V1.2.0; 2019-04; [A1.0]
AURIX™ TC39x Appendix to User's Manual	V1.0.0; 2018-08; [R1.0] V1.2.0; 2019-04; [A1.0]
AURIX™ TC3xx ED Target Specification ¹	V2.5.1; 2018-04; [R1.0] V2.8.0; 2019-05; [A1.0]
TC39x_DS_v10 TC39xB_DS_v11	V1.0; 2018-11; [R1.0] V1.1; 2019-09; [A1.0]
TC39x_Data_Sheet_Addendum	V1.1; 2018-08; [R1.0] V1.4; 2020-01; [A1.0]
AURIX TC3xx Safety Manual	V1.03; 2019-01; [R1.0] V1.08; 2020-03; [A1.0]
TC39x_BC_Errata_Sheet_v1_0 TC39xBD_Errata_Sheet_v1_0	V1.0; 2018-12-14; [R1.0] V1.0; 2019-12-19; [A1.0]
AURIX_TC39x_FMEDA_v1.01_ext	V1.01_ext; 2018-12- 17; [R1.0], [A1.0]
CPU SBST User Manual	V1.0.2.1.1; 2018-12- 14; [R1.0] V1.0.3.1.0; 2019-03- 05; [A1.0]
SPU SBST User Manual	V1.0.0.1.14; 2019-01- 10; [R1.0] V1.0.1.1.10; 2019-02- 11; [A1.0]
A2G_AURIX Safety Analysis Summary Report	V1.04; 2019-01- 07; [R1.0] V1.06; 2019-07- 03; [A1.0]
TriCore TC1.6.2 Core Architecture Manual; Core Architecture (Vol. 1)	V1.0; 2017-01-11; [R1.0] V1.2.2; 2020-01-15; [A1.0]
TriCore TC1.6.2 Core Architecture Manual; Instruction Set (Vol. 2)	V1.0; 2017-01- 11; [R1.0] V1.2.2; 2020-01-15; [A1.0]

¹ Distribution under NDA, only relevant for tool development not for application development
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2 References

- [1] ISO 26262, Road Vehicles Functional Safety, Version 2011

Revision history

3 Revision history

Document version	Date	Description of changes
V1.0	2022-04	Initial release

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Document reference

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